

StreamSentinel

An AI assistant that assesses the condition of urban streams from a single photo

Project report

OneAquaHealth IEEE Global Hackathon 2026 · Track 3: AI-Supported Assessment

Miryame Bendaoud

Master's degree (M2) MAS, Data Science track · Aix-Marseille University

Mohamed Bendaoud

Integrated preparatory programme (PeiP) · Polytech Marseille, Aix-Marseille University

Marseille, October 2026

1. Summary

StreamSentinel is an AI assistant that helps citizens and managers assess the condition of an urban stream from a single photo. On photos it has never seen, it pre-fills 61% of the OneAquaHealth observation form, with 71% of answers correct. It also estimates a health risk level, separately for humans and for animals, by combining the photo with weather data and river flow.

The AI never decides alone. The citizen reviews and corrects every suggestion, and any high risk is examined by a manager before any alert is issued.

The prototype combines five AI models (SegFormer, CLIP, YOLO11s fine-tuned by the team, OWLv2, Qwen2.5-VL-3B), three public data sources (Open-Meteo, Hub'Eau, OneAquaHealth's ENORA API), a web application with three spaces, and a FHIR R4 export that uses the OneAquaHealth app's codes. Each model was evaluated on independent images, and the results are reported with their confidence intervals.

Project built for the OneAquaHealth IEEE Global Hackathon 2026, Track 3: AI-Supported Assessment.

2. Context and challenges

OneAquaHealth is a European project that links the health of urban streams to the health of humans and animals, following the One Health approach. It monitors around a hundred sites in five cities (Coimbra, Toulouse, Benevento, Ghent and Oslo) and relies on citizen science: residents observe a stream and describe it in a mobile app.

The app's form has 9 steps: media (upstream, downstream, context and biodiversity photos, plus a short video), channel shape and banks, water appearance and flow, vegetation on both banks, habitats and natural debris, feelings, and finally an overall assessment in three categories. It is long, and answers vary with each person's experience. Observations are therefore uneven, sometimes wrong, and rarely turned into decisions.

Track 3 asks teams to use AI responsibly to help assess streams, without replacing human judgement. It expects validation checks, explainable AI and a human in the loop. The jury scores impact and alignment with the OneAquaHealth mission (30%), innovation (20%), technical implementation (20%), usability (15%), and feasibility and scalability (15%).

From this, we set three requirements:

- **Reliability:** the photo must be usable, and the AI must say when it is not sure.
- **Usefulness:** the output must fit OneAquaHealth's existing questionnaire and data formats.
- **Human control:** no alert is sent without validation by a person.

3. Approach

The project followed six phases, each validated before the next, with steps back whenever the evaluation justified it.

1. **Understanding the challenge.** Study of the rules and scoring grid, exploration of the OneAquaHealth app, and field-by-field reconstruction of its form with its internal codes.
2. **Image analysis pipeline.** Quality check, scene validation, reading of the water and banks, object detection, and fine-tuning of a litter detector.
3. **Form pre-filling.** Choice of an open-source vision-language model, prompt writing, handling of uncertainty and ambiguous cases.
4. **Context and risk.** Connection to weather and river flow data, writing of the risk rules, then linking each rule to the OneAquaHealth indicator factsheets.
5. **Application and integration.** Citizen and manager interface, live analysis API, FHIR export with the app's codes.
6. **Independent evaluation and fixes.** Building and annotating new test sets, measuring with confidence intervals, then removing or fixing what was not reliable.

The code, notebooks, evaluation sets and documentation are published in a public GitHub repository.

4. Proposed solution

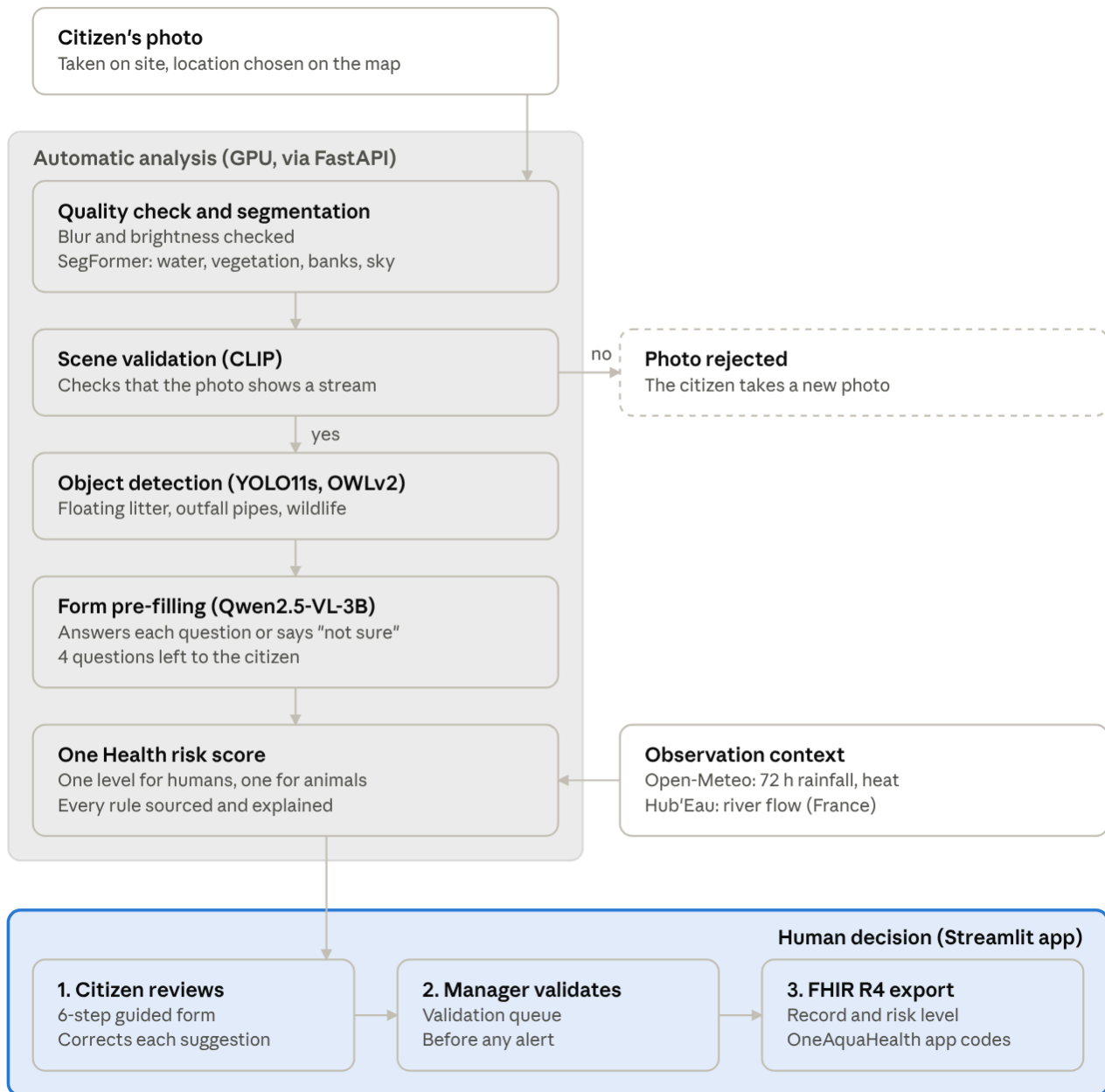
StreamSentinel turns a stream photo into a pre-filled observation record and an explained risk level, which the citizen and then the manager validate. Seven principles guided the design:

1. **Pretrained models, only one fine-tuned.** The pipeline combines existing models and image processing; only the litter detector was fine-tuned, because no suitable public model existed. This is a realistic choice for a hackathon, without a large annotated dataset.
2. **The AI can answer "not sure".** When the clues contradict each other, the question goes back to the citizen instead of being guessed.
3. **Every suggestion is explained.** The citizen sees where each answer comes from: proposed by the AI (with its confidence and reason), changed by the citizen, or left for the citizen to fill in.
4. **Some answers stay human.** The citizen's feelings are never pre-filled, and questions where the AI has not proven itself are left to the citizen.
5. **Every risk rule is sourced.** It refers to the OneAquaHealth indicator factsheets, or it is presented as a team assumption.
6. **No automatic alert.** A high risk sends the record to a manager, never directly to the public.
7. **Data stays compatible.** Answers are exported in FHIR R4 with the OneAquaHealth app's codes.

5. Architecture

StreamSentinel has two parts: an automatic analysis that runs on a GPU behind a FastAPI API, and a Streamlit web application where the citizen and then the manager make the decisions.

AI suggests, humans decide: from photo to FHIR export



StreamSentinel architecture · 5 analysis steps, 3 human decision steps

A photo rejected by the scene check goes no further, and the citizen receives instructions to retake it. All other photos are reviewed by the citizen before reaching the manager. Each component is developed and tested separately, so one can be replaced (for example by a larger model) without touching the others.

6. Analysis pipeline

Each step produces a readable, explainable output that the next step reuses. Only one model was trained by the team: the litter detector, because no suitable public model existed.

Step	Role	Method	Key choices
Quality check and scene reading	Discard unusable photos, then describe the scene and the water	Laplacian variance (blur) and brightness; SegFormer (ADE20K) for water, vegetation, buildings, ground and sky; colour (HSV) and texture analysis of water pixels	Bank type read from the ring of pixels around the water; shading estimated from tree canopy; turbidity, green water, foam and iridescence measured on water pixels only
Scene validation	Check that the photo shows a stream and matches the citizen's description	Zero-shot CLIP, against trap scenes (swimming pool, sea, street, car park, portrait)	Specific instructions for each rejection reason; an element described but missing from the photo triggers a request for a second photo
Object detection	Locate litter and objects without a dedicated dataset	YOLO11s fine-tuned on 2,058 floating litter images; zero-shot OWLv2 for outfall pipes, wildlife and sanitary debris	OWLv2 detections filtered (thresholds 0.30 to 0.35, boxes larger than 25% of the image rejected), down from 18–33 to 1–8 per photo; cross-checked with the water mask
Form pre-filling	Answer the OneAquaHealth form questions	Qwen2.5-VL-3B, an open-source vision-language model that runs on a T4 GPU; it receives the photo and the detections and answers in JSON with the app's codes	"Not sure" answer allowed; arbitration between pollution foam and white water; double check before reporting a dead animal; banks swapped depending on whether the photo faces upstream or downstream

Step	Role	Method	Key choices
Risk score	Estimate the risk for humans and for animals	Explicit point-based rules, using weather and river flow	Two separate scores; safety advice; 3-day trend; manager validation above a threshold

The models are loaded once on a T4 GPU (Google Colab) and exposed through a FastAPI API, which the app calls for each new photo. A version hosted on Hugging Face Spaces, with an on-demand GPU, was prepared to make the analysis available without Colab.

7. External data sources

A photo alone is not enough to judge a health risk: StreamSentinel combines it with the context of the place and time, using free public APIs.

Source	Data used	Role in the tool
Open-Meteo	Rainfall over the last 72 hours and today, rainfall forecast over 3 days, maximum temperature	After heavy rain, runoff and sewer overflows bring pollutants that are invisible on the photo; heat favours algal blooms. No API key needed.
Hub'Eau (hydrometry)	Flow measured at the nearest station, compared with its 30-day median	Very high flow signals a flood and a dangerous current; very low flow, more concentrated pollution. Data available in France only.
OneAquaHealth ENORA API	The project's research sites	Shown on an interactive map; the citizen places the observation on it.
Roboflow Universe	Floating litter datasets	Fine-tuning of the litter detector, then testing on an external set it had never seen.

Example from our tests in Paris: the Seine station at Paris-Austerlitz, 1.6 km away, measured 76.7 m³/s against a median of 81 m³/s, i.e. normal flow, consistent with the absence of rain recorded by Open-Meteo. A simulated flood raises the human risk by one or two levels, which confirms that the context does change the result.

8. One Health risk score

The score produces two separate levels, one for humans and one for animals, because the same observation does not expose them in the same way: a dog drinks and swims, and cyanobacteria are especially dangerous to it. Each triggered rule adds points; a total of 0 to 1 means low risk, 2 to 3 moderate risk, and 4 or more high risk.

Triggered rule	Humans	Animals	Justification
Visible sewage discharge	+3	+3	Factsheets VII, VIII and IX: enteric pathogens and antibiotic resistance genes
Dead animal reported (to be confirmed)	+3	+3	Factsheets III and I: transmissible pathogens, possible sign of a toxic bloom
Abnormal water colour	+2	+2	Factsheet I: algal blooms produce toxins
Foam on the water	+2	+2	Team assumption
Rain ≥ 20 mm in 72 h (≥ 10 mm: +1)	+2	+2	Factsheets VIII and IX: runoff increases pathogens; team thresholds
Flow $\geq 2\times$ the median ($\geq 1.5\times$: +1 for humans)	+2	+1	Team assumption: flood and dangerous current
Coloured water and heat ≥ 25 °C	+1	+2	Team assumption: cyanobacteria more likely
Turbid water	+1	+1	Team assumption
Visible drainage pipe	+1	+1	Factsheet IX: spread through urban runoff
Litter detected	+1	+1	Team assumption: injury or ingestion
Fast current after rain	+1	0	Team assumption: risk of falling or being swept away
Stagnant, warm water	0	+1	Team assumption
Flow $\leq 0.3\times$ the median	0	+1	Team assumption: more concentrated pollution

The factsheets cited are the OneAquaHealth Key Indicators factsheets (Schmeller et al., 2026). They explain why a signal matters for health, but give no thresholds for photo-based observations: the points and thresholds are therefore team choices, shown as such.

Each level comes with matching advice, for example "Avoid contact with the water and wash your hands after your visit" for a moderate human risk, or "Keep your dog from drinking or swimming" for a moderate animal risk. A 3-day trend switches to "rising" when more than 20 mm of rain is forecast. As soon as either level is high, the record goes to the manager's queue instead of triggering an alert. A "Why this level?" table shows the citizen and the manager every triggered rule, its points and its justification.

9. Application and integration

The Streamlit app, fully in English for international use, has three spaces, one for each audience.

Space	For whom	What happens there
Report a stream	Citizen	Choose a photo and place the observation on the map, follow a 6-step guided form modelled on the OneAquaHealth app, review the AI's suggestions, see the risk level and export the record
Validation queue	Manager	Review observations, read the "Why this level?" table, approve or reject, then export the validated record
How the AI decides	Everyone	Understand the analysis pipeline, the evaluation results and the limitations

Transparent suggestions. The annotated photo stays visible throughout the form. A label under each answer shows its origin: proposed by the AI (with its confidence and reason), changed by the citizen, or left for the citizen to fill in. Follow-up questions, such as vegetation type, only appear when the OneAquaHealth app would ask them.

Demo mode. Photos that have already been analysed let anyone use the app without a GPU. This is the version deployed online on Streamlit Community Cloud. When the citizen changes an answer, the risk is recalculated live. A map shows the OneAquaHealth research sites, and the selected photo places the observation on it automatically.

Integration with OneAquaHealth. The record is exported in FHIR R4, the standard for exchanging health data, at the end of the citizen form and again after the manager's decision. Each answer carries the code used by the OneAquaHealth app (for example

U_SHAPE becomes U, SLOW becomes NOR, FOAM becomes FO), and follow-up questions are only exported when the app would ask them. The FHIR validator found no structural errors.

10. Evaluation and results

Each model was tested on images that were used neither to tune it nor to select it. This independent evaluation replaced a first figure of 86.7%, which was too optimistic because it was measured on the photos used during development.

Protocol. For pre-filling, we built a set of 21 new Wikimedia Commons photos, then annotated every question by hand following the OneAquaHealth app's guidelines (banks defined looking downstream, "not sure" when the photo does not allow a decision). The AI's answers were compared with this ground truth. The scene check was tested on 66 photos, mixing rivers and off-topic images, with the threshold unchanged. The litter detector was measured twice: on the validation split of its fine-tuning set, then on an external set it had never seen.

Step	Test set	Metric	Result [95% CI]
Form pre-filling (Qwen2.5-VL-3B)	21 new Wikimedia photos, hand-annotated	Correct answers	71% [62–80%]
Form pre-filling (Qwen2.5-VL-3B)	Same 21 photos	Share of questions pre-filled	61% [45–73%]
Scene check (CLIP)	66 new photos	Streams accepted (sensitivity)	92% [79–97%]
Scene check (CLIP)	Same 66 photos	Off-topic images rejected (specificity)	79% [62–90%]
Litter detector (YOLO11s)	599 validation images from the fine-tuning set	Precision / recall / mAP50	0.85 / 0.75 / 0.80
Litter detector (YOLO11s)	Floating Trash Detection: 382 unseen images	Precision / recall at threshold 0.25	0.77 / 0.43
Litter detector (YOLO11s)	Same 382 images	F1 / mAP50	0.51 [0.47–0.55] / 0.47
FHIR R4 export	Exported records	Structural errors from the validator	0

Confidence intervals are obtained by bootstrapping over photos. They are wide: these results are a first estimate, not a validation.

Reading the results. The scene check is a good first filter: a photo wrongly accepted is then reviewed by the citizen, and by the manager for serious cases. The litter detector performs very well on images close to its training data, but its performance drops clearly on an external set. This is the expected gap between internal validation and generalisation, which is why we report both. Lowering its threshold barely helps (best F1 of 0.557 against 0.548), so the 0.25 threshold is kept. The first scene-reading measurements (90.9% on bank type, for example) were obtained on the 11 photos used to set the thresholds; they are indicative only and are not included in the table.

11. Challenges and solutions

For each challenge, we chose to remove or limit what was not reliable rather than display it.

Challenge	Solution
A first figure of 86.7% correct answers, measured on 8 photos already used during development	New set of 21 unseen photos, hand-annotated, with confidence intervals: 71%
Channel shape, vegetation on both banks and overall condition: accuracy no better than chance	Pre-filling disabled for these 4 questions; the citizen answers them alone
The public dead-fish model (advertised mAP50 of 88.8%) flagged a guard rail and clouds	Model removed; detection handed to the vision-language model, with a double check
The vision-language model mistook bottles for dead fish; a list of objects to exclude made it worse (from 3 to 5 false positives)	Prompt rewritten around positive anatomical signs: 1 false positive
OWLv2 produced 18 to 33 detections per photo, including boxes covering the whole scene	Per-class thresholds, object descriptions instead of scene descriptions, oversized boxes rejected: 1 to 8 detections
Muddy water was not detected	Tighter turbidity thresholds
The "not sure" answer was rejected as invalid (66 error messages)	"Not sure" accepted as a valid answer

Challenge	Solution
Left and right banks judged with different rules, while the app defines them looking downstream	Same rule for both banks, swapped according to the photo's direction
No GPU on our computers	Models on Colab, exposed through a FastAPI API and an ngrok tunnel; offline demo mode with pre-analysed photos
OpenCV 5 removed the face detection tool planned for blurring	Blurring kept in the architecture, documented as a limitation
Incompatible libraries (inference-sdk with Python 3.13, new transformers versions for CLIP)	Direct call to the Roboflow API; adapter function for CLIP outputs

12. Limitations and ethics

Known limitations are documented rather than hidden.

- **Small test sets.** 21 and 66 photos: confidence intervals are wide, and the results are a first estimate.
- **A single annotator.** The form test set was annotated by one person; inter-annotator agreement could not be measured.
- **Unrealistic photos.** Wikimedia photos are sharper and more varied than real phone photos taken by citizens.
- **Incomplete litter detection.** On new images, the detector misses more than half of the litter.
- **Known confusions.** CLIP sometimes mistakes a wide, calm river for a lake, and lets some open landscapes such as the sea through. Waterfall spray can look like foam, and dark, shallow water like soil.
- **Language.** CLIP is trained in English: French descriptions go through a keyword table that does not cover the whole vocabulary.
- **Flow data limited to France.** In the other OneAquaHealth cities, the risk relies on weather alone.
- **Unvalidated risk thresholds.** They are sourced where possible, but remain team choices.
- **Blurring not active.** Blurring of faces and licence plates (GDPR) is planned before any analysis, but not yet enabled.

- **Demo deployment.** Live analysis depends on a Colab GPU reached through an ngrok tunnel: suitable for a demo, not for real use.

Ethically, the central principle is human decision-making: the AI suggests, the citizen corrects, and the manager validates before any alert.

13. Next steps

The priority is to test the tool on real citizen photos, taken by phone at OneAquaHealth sites.

1. **Validate in the field.** Build a test set of real citizen photos, annotated by at least two people, to measure inter-annotator agreement.
2. **Improve the models.** Re-evaluate the 4 disabled questions with a larger vision-language model, and fine-tune the litter detector on urban stream photos to improve its recall.
3. **Have the score validated.** Submit the points and thresholds to OneAquaHealth experts, and add a rule on invasive plants, which factsheet XI links to allergenic pollen and irritation.
4. **Recognise species.** Connect Pl@ntNet to identify invasive plants on the biodiversity photo.
5. **Protect privacy.** Enable face and licence plate blurring before any public use.
6. **Scale up.** Host the models on a stable GPU (for example a Hugging Face Space with an on-demand GPU), offer a mobile version with guided capture, and integrate the FHIR export into the OneAquaHealth platform.
7. **Extend beyond France.** Add flow data sources for the project's other cities.

14. Team, deliverables and references

Team. Miryame and Mohamed Bendaoud. Source code: GitHub repository `momo25bend/streamsentrinel`.

Deliverables. Source code, notebooks, evaluation sets and English documentation in the public GitHub repository (github.com/momo25bend/streamsentrinel); online app in demo mode (streamsentrinel.streamlit.app); demo video on the project's Devpost page.

Data.

- Test photos: Wikimedia Commons, authors and licences listed in `credits_photos.csv`.
- Litter detector fine-tuning: Roboflow Universe, floating-waste dataset from RF100-VL (MIT licence).

- External detector test: Roboflow Universe, Floating Trash Detection (CC BY 4.0), never used for training.
- Research sites: OneAquaHealth ENORA API.
- Context: Open-Meteo (weather) and Hub'Eau (French river flow).

Reference. Schmeller et al. (2026). *OneAquaHealth Key Indicators of Ecosystem and Biological Health – Factsheets*. [doi:10.5281/zenodo.20345207](https://doi.org/10.5281/zenodo.20345207), CC BY 4.0.